ULUBILGE ULUSOY, PH.D.

Human factors engineer specializing in AI-enabled crew support. Dedicated to designing, evaluating, and advancing human-centered systems in high-risk, high-stakes environments. Skilled in leveraging experimental testing and data-driven insights to optimize performance, safety, and mission success.

CONTACT

- ulubilgeulusoy@gmail.com
- (323)-690-0983
- 5863 Arapahoe Ave, Boulder, CO.

- Portfolio
- GitHub Profile
- LinkedIn Profile
- Publications

EDUCATION

Ph.D., Astronautical Engineering
University of Southern California (USC)
GPA: 3.90/4.00 • Year: 2025

M.S., Astronautical Engineering
University of Southern California (USC)
GPA: 3.92/4.00 • Year: 2020

B.S., Astronautical Engineering
Istanbul Technical University (ITU)
GPA: 3.70/4.00 • Year: 2018

EXPERTISE

Human Factors Engineering
Human–AI Interaction
Human Subjects Research
Human-in-the-Loop Testing
IRB Protocol Management
Simulated Crew Operations and Training
Human–Robot Interaction
AI-Augmented Development

AWARDS & HONORS

Young Pioneer Award, Finalist, IAF (2024)
Best Research Assistant, USC (2022)
Rocket Scientist of the Year, USC (2020)

GRANTS & FELLOWSHIPS

NASA STRI, HOME (2021–2025)
USC Viterbi Grad Fellowship (2020–2021)

STACK

Python (OpenCV, PyQt, Tkinter)
C++ (ViSP)
BIOPAC
Lab Streaming Layer
MATLAB & R
Qualtrics
LaTeX
Open Broadcaster Software

LANGUAGE

English (Fluent)
Turkish (Native)

PROFESSIONAL EXPERIENCE

07/2025 - Present
University of Colorado Boulder — Postdoctoral Associate

- Leading end-to-end experimental protocol design and execution for a human-robot interaction study investigating the utilization of robots to support astronauts during maintenance operations.
- Developed Franka Research 3 robot arm capabilities using AI-augmented (generative AI-assisted) development for kinesthetic teaching in Python and visual servoing (ViSP) in C++, incorporating safety features for real-time human interaction in co-located task execution.
- Developed a GUI (Tkinter) in Python for remote controlling the FR3 capabilities (moving arm and gripper by using kinesthetic teaching and visual servoing) via X-server (X11).
- Established data collection and processing pipelines for physiological (BIOPAC ECG and respiration), computer vision (facial impressions), and survey-based (Qualtrics) measures.
- Developed Lab Streaming Layer (LSL) applications in Python to synchronize physiological, computer vision, and behavioral data streams.
- Developed GUIs (Tkinter) in Python for data visualization, as well as data analysis scripts for the collected data.
- Designed an ecologically valid space habitat maintenance scenario with associated experimental layouts, session workflows, and participant training protocols.

08/2020 - 05/2025
University of Southern California — Research Assistant

- Conducted human factors research funded by NASA and advised by former NASA astronaut Professor Garrett E. Reisman for 5-years on integrating AI into crew operations.
- Led end-to-end experimental protocol design, IRB process management and experiment execution in two separate occasions for human-AI interaction studies involving 60 participants.
- Designed experimental layouts, session workflows, and participant training protocols for these experiments and trained 60 participants to perform a simulated space habitat maintenance task during experimental sessions.
- Rehearsed training and operational protocols for experiments with former NASA astronaut Garrett Reisman.
- Built computer vision-based applications (OpenCV) in Python for data collection; performed statistical analysis in MATLAB and R.
- Conducted a survey study with ten former astronauts to characterize interaction dynamics between astronauts and mission control during maintenance operations, informing AI integration into crew operations.
- Performed comprehensive literature reviews on human factors metrics and human spaceflight operational and training procedures.

11/2019 - 11/2020
USC Liquid Propulsion Laboratory — Lead Engineer

- Managed a nonprofit academic propulsion group, leading 40+ students and overseeing an annual budget of approximately $60k.
- Led statement-of-work development and coordinating with University Corporate Relations to launch the sponsored research collaboration.
- Managed end-to-end project execution for the design of a liquid rocket engine test stand in collaboration with Pangea Propulsion, serving as project coordinator overseeing timelines, deliverables, and technical alignment.
- Implemented industry-aligned organizational and technical operating standards across laboratory activities.
- Coordinated with USC Environmental Health & Safety (EH&S) to establish testing protocols and COVID-era operational procedures.
- Defined high-level project goals for the design and testing of rocket engines and feed systems.